Power BI Assignment 1 – Retail Sales Analytics (Easy)

Oct 13, 2026 · @Dr. Subhabaha Pal

Overview

You will build a five-page Power BI report (20 visuals) for a national retail chain from a ready-made star-schema dataset of about 24,000 sales transactions.

Item	Details
Institution	IILM Jaipur
Programme	MBA
Assignment	Power BI Assignment 1 of 2
Difficulty	Easy
Tool	Power BI Desktop (latest version)
Dataset file	ShopSphere_Retail_StarSchema.xlsx
Work mode	Individual
Marks	50
Submission date	To be announced by the faculty

Learning objectives

By the end of this assignment you should be able to:

1. Import a multi-sheet Excel workbook into Power BI and check data types.
2. Build a star-schema data model with one-to-many relationships.
3. Write basic DAX measures (SUM, DIVIDE, DISTINCTCOUNT, CALCULATE, TOTALYTD).
4. Choose the right visual for a business question and format it clearly.
5. Combine visuals, slicers and KPI cards into dashboards that tell a story.
6. Write short, data-backed business insights.

Business scenario

ShopSphere Retail is a fictitious Indian chain with 24 stores across four regions. It sells Electronics, Fashion, Home & Kitchen, Grocery and Beauty & Personal Care. Management has two years of sales data (January 2024 to December 2025) and wants a self-service dashboard to answer four questions:

- Is the business growing, and how strong is seasonality?
- Which categories, products and brands drive revenue and profit?
- Which stores and regions over- or under-perform?

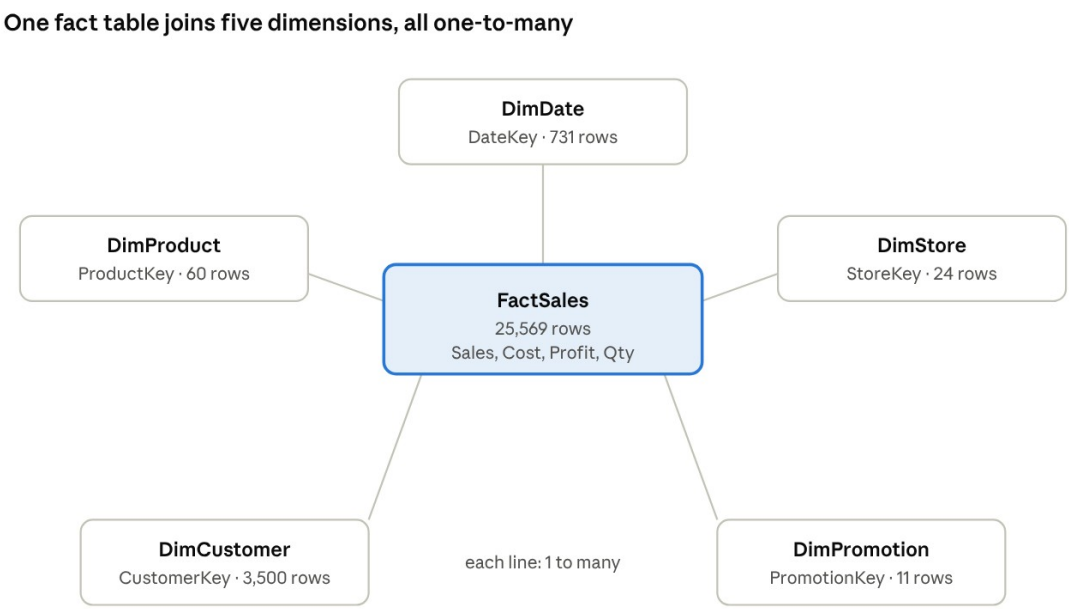
- Who are our customers, and do promotions actually pay off?

You are the business analyst asked to build this report.

Dataset and star schema

The workbook ShopSphere_Retail_StarSchema.xlsx has one fact table (25,569 rows) and five dimension tables, plus a ReadMe sheet. Every sheet is already formatted as an Excel table with the same name. All values are in Indian Rupees (INR), and the data is synthetic.

Sheet	Type	Rows	Key column	Main columns
FactSales	Fact	25,569	SalesID	DateKey, ProductKey, StoreKey, CustomerKey, PromotionKey, PaymentMode, Quantity, UnitPrice, DiscountPct, SalesAmount, CostAmount, Profit
DimDate	Dimension	731	DateKey	Date, Year, Quarter, MonthNo, MonthName, MonthYear, MonthYearSort, DayName, DayType, FinancialYear
DimProduct	Dimension	60	ProductKey	ProductName, Category, SubCategory, Brand, UnitPrice, UnitCost
DimStore	Dimension	24	StoreKey	StoreName, StoreType, City, State, Region, OpeningDate, StoreSizeSqFt
DimCustomer	Dimension	3,500	CustomerKey	CustomerName, Gender, AgeGroup, Segment, City, Region, LoyaltyMember
DimPromotion	Dimension	11	PromotionKey	PromotionName, PromotionType, DiscountPct, StartDate, EndDate



ShopSphere star schema · 1 fact, 5 dimensions

Every dimension filters FactSales through its Key column; your Power BI model view should look like this.

Column notes:

- SalesAmount is net of discount (Quantity × UnitPrice × (1 − DiscountPct)).
- Profit = SalesAmount − CostAmount. Some promotional lines have negative profit, which is intentional.
- DateKey is a whole number in yyyymmdd format (for example 20250315).
- FinancialYear follows the Indian April–March year (for example FY24-25).
- Two stores (Surat and Patna) opened on 1 July 2024, so they have no sales before that date.

Part A – Data preparation and modelling (10 marks)

Complete these steps before you build any visual.

1. **Load data.** Home → Get Data → Excel Workbook → select all six tables (not the ReadMe sheet) → Transform Data.
2. **Check data types in Power Query.** Date, OpeningDate, StartDate and EndDate = Date. DiscountPct = Percentage. SalesAmount, CostAmount, Profit, UnitPrice, UnitCost = Fixed decimal (currency). All Key columns = Whole number.
3. **Build relationships** in Model view. Each dimension relates one-to-many (1:*) to FactSales, single cross-filter direction:
    a. DimDate[DateKey] → FactSales[DateKey]
    b. DimProduct[ProductKey] → FactSales[ProductKey]
    c. DimStore[StoreKey] → FactSales[StoreKey]
    d. DimCustomer[CustomerKey] → FactSales[CustomerKey]

e. DimPromotion[PromotionKey] → FactSales[PromotionKey]

4. **Mark DimDate as the date table** (Table tools → Mark as date table → Date).
5. **Sort columns:** MonthName by MonthNo, and MonthYear by MonthYearSort.
6. **Create a hierarchy** in DimDate: Year → Quarter → MonthName → Date. Create a second hierarchy in DimProduct: Category → SubCategory → ProductName.
7. **Hide key columns** in report view (all Key columns in FactSales) so report users see only meaningful fields.
8. **Take a screenshot** of the finished model for your submission.

Part B – DAX measures (10 marks)

Create a table called _Measures (Enter data → blank table) and store every measure there. Format currency measures with the ₹ symbol and no decimals.

#	Measure	Suggested DAX
1	Total Sales	`SUM ( FactSales[SalesAmount] )`
2	Total Cost	`SUM ( FactSales[CostAmount] )`
3	Total Profit	`SUM ( FactSales[Profit] )`
4	Profit Margin %	`DIVIDE ( [Total Profit], [Total Sales] )`
5	Total Quantity	`SUM ( FactSales[Quantity] )`
6	No. of Transactions	`COUNTROWS ( FactSales )`
7	Unique Customers	`DISTINCTCOUNT ( FactSales[CustomerKey] )`
8	Avg Transaction Value	`DIVIDE ( [Total Sales], [No. of Transactions] )`
9	Sales YTD	`TOTALYTD ( [Total Sales], DimDate[Date] )`
10	Sales LY	`CALCULATE ( [Total Sales], SAMEPERIODLASTYEAR ( DimDate[Date] ) )`
11	YoY Growth %	`DIVIDE ( [Total Sales] - [Sales LY], [Sales LY] )`
12	Promo Sales	`CALCULATE ( [Total Sales], DimPromotion[PromotionType] <> "None" )`
13	Promo Sales %	`DIVIDE ( [Promo Sales], [Total Sales] )`
14	Sales per Sq Ft	`DIVIDE ( [Total Sales], SUM ( DimStore[StoreSizeSqFt] ) )`

Also add one calculated column, DimProduct[Price Band] = "Budget" (below ₹1,000), "Mid" (₹1,000–10,000) or "Premium" (above ₹10,000), using IF or SWITCH.

Part C – Visualizations and dashboards (20 marks)

Build five report pages (dashboards). Each page has exactly four visuals plus the slicers listed, so the report has 20 visuals in total. Give every visual a clear title that states the question it answers, and use one consistent colour theme.

Dashboard 1 – Executive Overview

Slicers: Year, Region.

#	Visual type	Fields	Business question
V1	Multi-row card (KPI strip)	Total Sales, Total Profit, Profit Margin %, No. of Transactions, Unique Customers	What are our headline numbers?
V2	Line chart	X: DimDate[MonthYear]; Y: Total Sales	How do sales move month by month over two years?
V3	Clustered column chart	X: MonthName; Y: Total Sales; Legend: Year	Is 2025 ahead of 2024 in every month?
V4	Donut chart	Legend: Region; Values: Total Sales	Which region contributes the most revenue?

Dashboard 2 – Product and Category Performance

Slicers: Year, Category.

#	Visual type	Fields	Business question
V5	Clustered bar chart	Y: Category; X: Total Sales and Total Profit	Which categories earn revenue vs. profit?
V6	Treemap	Category → SubCategory; Values: Total Sales	How is revenue split inside each category?
V7	Bar chart with Top N filter	Y: ProductName; X: Total Sales; Filter: Top 10 by Total Sales	Which 10 products sell the most?
V8	Scatter chart	Details: ProductName; X: Total Sales; Y: Profit Margin %; Size: Total Quantity; Legend: Category	Which products are high-volume but low-margin?

Dashboard 3 – Store and Regional Performance

Slicers: Year, StoreType.

#	Visual type	Fields	Business question
V9	Map (bubble)	Location: DimStore[City]; Bubble size: Total Sales	Where are our strongest cities?
V10	Bar chart, sorted descending, with data bars or colour by value	Y: StoreName; X: Total Sales	Which stores are the top and bottom performers?
V11	Matrix with conditional formatting	Rows: Region → State → StoreName; Values: Total Sales, Total Profit, Profit Margin %, YoY Growth %	How does each store compare on growth and margin?
V12	Clustered column chart	X: StoreType; Y: Sales per Sq Ft	Which store format uses its space best?

Dashboard 4 – Customer Insights

Slicers: Year, Region, LoyaltyMember.

#	Visual type	Fields	Business question
V13	Clustered column chart	X: AgeGroup; Y: Total Sales; Legend: Gender	Which age group and gender buy the most?
V14	Pie chart	Legend: Segment; Values: Total Sales	How much revenue comes from Premium, Regular and New customers?
V15	Bar chart	Y: Segment; X: Avg Transaction Value	Do Premium customers spend more per bill?
V16	100% stacked column chart	X: Year; Y: Total Sales; Legend: PaymentMode	Is the payment mix shifting (UPI vs. cash)?

Dashboard 5 – Promotions and Time Analysis

Slicers: Year, Category.

#	Visual type	Fields	Business question
V17	Line and clustered column chart (combo)	X: PromotionName; Columns: Total Sales; Line: Profit Margin %	Which promotions bring sales but hurt margin?
V18	Line chart	X: MonthName; Y: Sales YTD; Legend: Year	How does cumulative sales in 2025 track against 2024?
V19	Column chart	X: DayName (sorted by	Which days of the week are

#	Visual type	Fields	Business question
		DayOfWeekNo); Y: Total Sales	busiest?
V20	KPI visual	Value: Total Sales; Trend axis: MonthYear; Target: Sales LY	Are we beating last year's sales?

Formatting checklist

- Add a page title and the company name on each page.
- Use the same slicer position (top or left) on every page.
- Show data labels where values are compared, and remove clutter (gridlines, extra decimals).
- Use conditional formatting in V10 and V11 (red for low, green for high).
- Add page navigation buttons from Dashboard 1 to the other four pages.

Part D – Business insights (10 marks)

Answer each question in two to four sentences in a short Word or PDF report. Quote the numbers from your dashboard and name the visual (V1–V20) that supports each answer.

1. By what percentage did total sales grow from 2024 to 2025? Which month is the peak month and why might that be?
2. Which category contributes the most revenue, and which earns the highest profit margin? What does this mean for shelf-space decisions?
3. Which category grew the fastest between 2024 and 2025, and which category declined?
4. About how many of the 60 products make up 80% of sales? (Hint: sort V7 or build a Pareto view.)
5. Name the three weakest stores. Is the problem the store format, the region, or the individual store?
6. Which age group spends the most? How much of revenue comes from Premium customers, compared with their share of the customer base?
7. How has the share of UPI and cash payments changed between 2024 and 2025?
8. Which promotion produced the lowest profit margin? Would you recommend running it again? Justify your answer.
9. Do weekends contribute more than their share of days (2 of 7)?
10. Write three recommendations for ShopSphere's management, each backed by one visual.

Submission guidelines

Submit one zip file named `RollNo_Name_PBI_A1.zip` on the LMS containing:

- The Power BI file (`.pbix`) with all five dashboards.
- A PDF export of the report (File → Export → Export to PDF).
- The insights report (Word or PDF, at most 4 pages) with the data model screenshot from Part A.

This is an individual assignment. Identical .pbix files or copied insights will receive zero marks.

Evaluation rubric (50 marks)

Component	Marks	What earns full marks
Part A – Data preparation and model	10	Correct data types, five 1:* relationships, date table marked, sorting and hierarchies done
Part B – DAX measures	10	All 14 measures and the calculated column return correct values and are formatted
Part C – 20 visuals in 5 dashboards	20	Right visual for each question, clear titles, working slicers, consistent theme (1 mark per visual)
Part D – Business insights	10	Accurate numbers, clear reasoning, practical recommendations
Total	**50**	